

Categorical Data Encoding Techniques in Machine Learning: Binary, Ordinal, Target, and Beyond

A Comprehensive Research Paper with Experimental Figures and Referenced Sources
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Abstract

*Machine learning algorithms require numerical input, yet real-world datasets routinely contain categorical features such as city names, product colours, or education levels. **Encoding** converts these categories into numeric representations while minimising bias, inflated dimensionality, and information loss. This paper provides a comprehensive review of six major encoding techniques—**Label, One-Hot, Ordinal, Binary, Target, and Frequency Encoding**—covering their mathematical formulation, advantages, disadvantages, and practical guidelines. Five original coloured figures illustrate key concepts. The paper concludes with a decision framework and references to peer-reviewed and industry sources.*

1. Introduction

Most machine learning algorithms—from linear regression to neural networks—operate exclusively on numerical matrices. Real-world data, however, is rich with categorical variables: colours, cities, job titles, education levels, and product types. Before any model can be trained, these must be converted into numbers without introducing spurious information. This process is called **categorical encoding**. The choice of encoding technique dramatically affects model accuracy, memory consumption, and generalisation. ^{[1][2]}

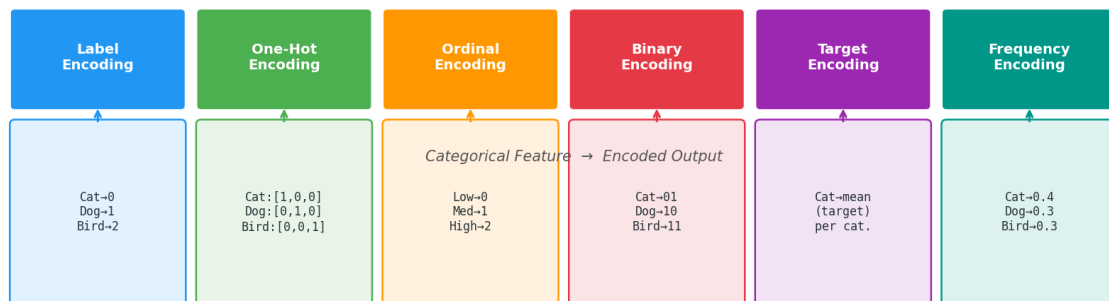
Encoding techniques differ along three critical axes: (1) whether they impose an **ordinal order** on categories, (2) the **dimensionality** they introduce, and (3) whether they incorporate **target information** (and thus carry overfitting risk). Choosing the wrong encoding can mislead linear models into learning false relationships or explode the feature space to thousands of columns. ^[2]

This paper systematically reviews the six most important encoding methods, provides step-by-step mathematical explanations, compares their properties in structured tables, and offers a practitioner decision guide.

2. Overview of Encoding Methods

Figure 1 gives a visual summary of all six encoding methods side by side, with colour-coded examples for a three-category feature {Cat, Dog, Bird}.

Figure 1 – Six Common Categorical Encoding Methods



Each method converts text categories → numbers for ML algorithms

Figure 1 – Six categorical encoding methods: name (header, coloured), example output (box below). Each converts text categories into numbers usable by ML algorithms.

The six methods occupy different positions in the trade-off space between dimensionality, ordinal assumptions, and target dependency:

- Label Encoding: One integer per category. Compact but implies ordinal order.
- One-Hot Encoding: One binary column per category. Order-free but high-dimensional.
- Ordinal Encoding: Integers assigned to preserve a known natural order.
- Binary Encoding: Integer → binary bits → one column per bit. Balances compactness and order-freedom.

- Target Encoding: Replace category with mean target value. High predictive power but overfitting risk.
- Frequency Encoding: Replace category with its relative frequency in the training set.

3. Label Encoding

Label encoding assigns a unique integer to each category in a single column. For a feature with categories $C = \{c_1, c_2, \dots, c_n\}$, the mapping is:

$$f(c_i) = i - 1 \quad (i = 1, 2, \dots, n)$$

3.1 Advantages

- Extremely memory efficient: replaces one column with one integer column regardless of cardinality.
- Compatible with tree-based models (decision trees, random forests, gradient boosting) that split on thresholds and are unaffected by ordinal assumptions.
- Simple and fast to compute.

3.2 Disadvantages

- Implies a false ordinal relationship for nominal categories: the model may learn that "Berlin" (2) > "Cairo" (0), which is meaningless. ^[2]
- Misleads linear models and neural networks that interpret numerical magnitude.
- Should never be used on nominal data with linear models.

4. One-Hot Encoding

One-Hot Encoding (OHE) creates n binary columns for a feature with n unique categories. For category c_i , column j gets value 1 if $i = j$, else 0:

$$\text{OHE}(c_i)_j = 1 \quad \text{if } i=j, \quad \text{else } 0 \quad (j = 1, \dots, n)$$

4.1 Advantages

- No ordinal assumption—all categories are treated as equidistant.
- Works correctly with linear models, logistic regression, and neural networks.
- Interpretable: each column corresponds directly to one category.

4.2 Disadvantages

- Dimensionality explosion: a feature with 1000 unique values creates 1000 binary columns. This is the "curse of dimensionality" in practice. ^[1]
- Creates multicollinearity (the dummy variable trap): one column is always predictable from the others; often one column is dropped.
- Memory-intensive for high-cardinality features (e.g. ZIP codes, user IDs).

5. Binary Encoding

Binary Encoding is a **two-step hybrid** that bridges the gap between Label Encoding (1 column, false order) and One-Hot Encoding (n columns, no order). It first applies label encoding, converts each integer to its binary representation, then splits the bits into separate columns. ^{[1][3]}

Step 1: $c_i \rightarrow \text{integer label } i$

Step 2: $i \rightarrow \text{binary string } b_1b_2\dots b_k \quad \text{where } k = \lceil \log_2(n+1) \rceil$

Step 3: each bit b_j becomes a separate column

For $n = 5$ categories, only $k = 3$ columns are needed instead of 5 (One-Hot). For $n = 1000$, binary encoding needs only $\lceil \log_2(1001) \rceil = 10$ columns versus 1000.

Figure 2 - Binary Encoding: Step-by-Step from Categories → Binary Bit Columns

Step 1: Original Categories		Step 2: Label Encoding (Integer)		Step 3: Binary Encoding (3 bit columns)					
				Category	Int	Binary	Bit2	Bit1	Bit0
Purple		Purple → 4		Red	0	000	0	0	0
Yellow		Yellow → 3		Blue	1	001	0	0	1
Green		Green → 2		Green	2	010	0	1	0
Blue		Blue → 1		Yellow	3	011	0	1	1
Red		Red → 0		Purple	4	100	1	0	0

Figure 2 – Binary Encoding step-by-step: from original categories (coloured bars) → label integers → 3-bit binary columns (table, right). Only 3 columns needed for 5 categories vs 5 for One-Hot.

5.1 Advantages

- Memory-efficient for high-cardinality features: $O(\log_2 n)$ columns vs $O(n)$ for One-Hot. ^[2]
- Avoids ordinal assumptions present in Label Encoding.
- Reduces the curse of dimensionality while retaining category distinctness.

5.2 Disadvantages

- Less interpretable than Label or One-Hot Encoding: individual bits have no semantic meaning. ^[2]
- Binary bit patterns can still imply partial ordinal relationships across bit boundaries.
- Requires careful handling of missing values during binary conversion.

6. Ordinal Encoding

Ordinal Encoding is appropriate when the categories have a **meaningful natural order**. The mapping must be domain-defined, not arbitrary. For example:

```
f("No Education")=0, f("High School")=1, f("Bachelor")=2,
f("Master")=3, f("PhD")=4
```

Unlike label encoding (where integers are assigned alphabetically or by first appearance), ordinal encoding **explicitly encodes rank information** that is semantically correct. [2]

Figure 3 - Ordinal Encoding (correct) vs Label Encoding on Nominal Data (incorrect)

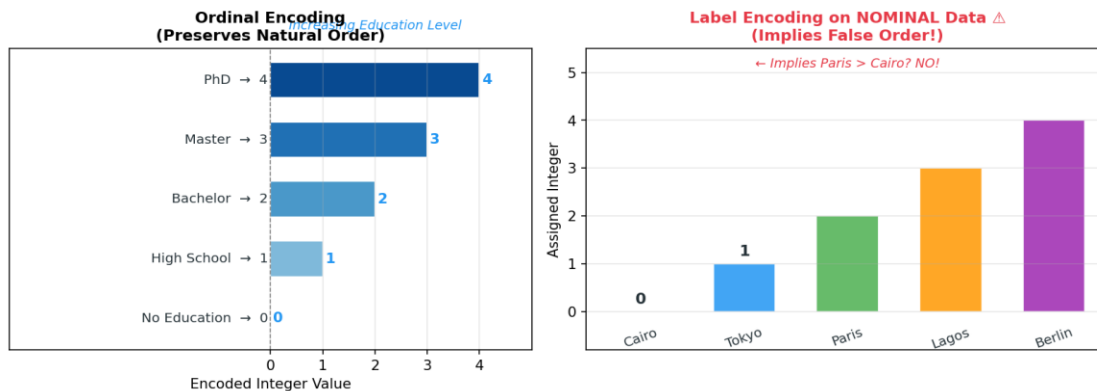


Figure 3 – Left: Ordinal encoding on education levels (correct—natural order preserved, blue gradient). Right: Label encoding on city names (incorrect—implies Paris > Cairo, shown in red warning). Never apply label encoding to nominal data in linear models.

6.1 Advantages

- Preserves meaningful hierarchical relationships — a regression model correctly learns that a PhD holder is "more educated" than a High School graduate. [2]
- Compact: single column regardless of cardinality.
- Works well with ordinal-aware models (regression, gradient boosting).

6.2 Disadvantages

- Unsuitable for nominal categories without natural order (colours, city names, product types): imposes false ordinal structure. [1]
- Implies equal spacing between ranks, which may not reflect true differences (e.g. distance from "High School" to "Bachelor" ≠ "Bachelor" to "PhD").
- Requires manual definition of the order mapping; cannot be inferred automatically.

7. Target Encoding

Target Encoding (also called **Mean Encoding**) replaces each category with the **mean of the target variable** for all observations belonging to that category:

$$TE(c_i) = E[Y | X = c_i] = (1/n_i) \sum y_j \quad \text{for all } j \text{ where } x_j = c_i$$

To prevent overfitting on rare categories (few observations → unreliable mean), **smoothing** blends the category mean with the global mean:

$$\text{TE_smooth}(c_i) = \lambda_i \cdot \text{mean}(Y|c_i) + (1-\lambda_i) \cdot \text{mean}(Y)$$

where $\lambda_i = n_i / (n_i + m)$ and m is a smoothing factor (often $m=300$). As $n_i \rightarrow \infty$, $\lambda_i \rightarrow 1$ (trust the category mean); as $n_i \rightarrow 0$, $\lambda_i \rightarrow 0$ (trust the global mean). ^{[2][4]}

Figure 4 - Target Encoding: Mean Encoding with Smoothing Against Overfitting

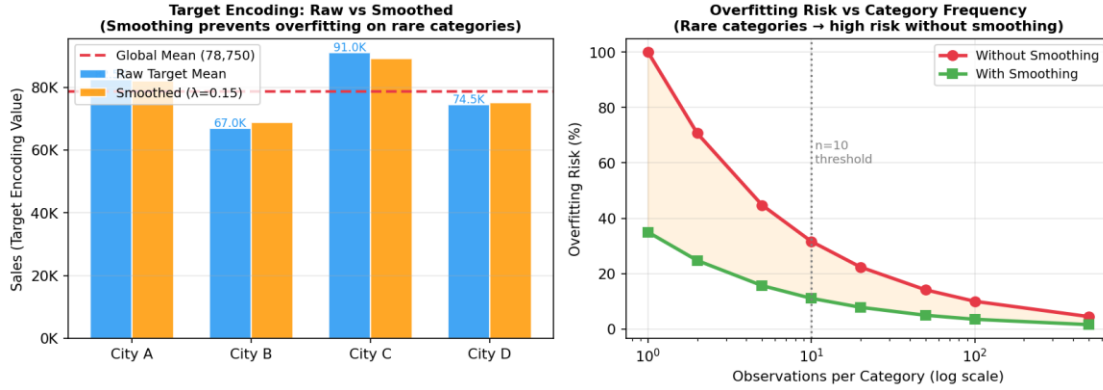


Figure 4 – Left: Raw vs smoothed target encoding values for 4 cities (red dashed line = global mean). Smoothing pulls rare categories toward the global mean. Right: Overfitting risk curve—without smoothing, rare categories (low n) carry very high variance.

7.1 Advantages

- Captures the statistical relationship between category and target: powerful for high-cardinality nominal features where One-Hot would be impractical. ^[2]
- Low dimensionality: replaces n columns (One-Hot) with a single column. ^[4]
- Ideal for features like ZIP code, user ID, or product SKU combined with a strong target signal.

7.2 Disadvantages

- High overfitting risk without cross-validation or smoothing: the encoding directly uses target information. ^[2]
- Data leakage if the encoding is fit on the full dataset including test data—must be computed only on training folds. ^[1]
- Fails with new categories at inference time (unseen at training) unless a fallback (global mean) is defined.

8. Frequency Encoding

Frequency Encoding replaces each category with its **relative frequency** in the training set:

$$\text{FE}(c_i) = \text{count}(c_i) / n_{\text{total}}$$

This is a **target-free** alternative to label encoding for nominal data, preserving the distributional importance of each category. Common categories receive higher values; rare ones receive lower. It is often used as a feature alongside other encodings. ^[2]

8.1 Advantages

- No ordinal assumption; encodes population rarity which can be informative.

- Single column, no dimensionality increase.
- Simple to compute and interpret.

8.2 Disadvantages

- Two categories with the same frequency get the same encoded value — collisions are possible.
- Frequency may not correlate with target relevance (rare $\neq$ unimportant).
- Sensitive to class imbalance and dataset size changes.

9. Comparative Analysis

9.1 Theoretical Comparison Table

Encoding	Output Type	Columns (n cats)	Ordinal Assumption	Overfit Risk	Best Use Case
Label	Integer	1	Yes (implicit)	Very Low	Tree models, ordinal data
One-Hot	Binary 0/1	n	None	Low	Linear models, low cardinality
Ordinal	Integer	1	Yes (explicit)	Very Low	Naturally ordered categories
Binary	Binary bits	$\log_2(n+1)$	None	Low	High-cardinality nominal
Target	Float	1	None	High	High-cardinality + strong target signal
Frequency	Float	1	None	Low	Any nominal when order is unknown

Table 1 – Theoretical property comparison of six encoding methods.

9.2 Dimensionality and Multi-Criteria Radar

Figure 5 - Dimensionality Growth and Multi-Criteria Quality Comparison Across Encoding Methods

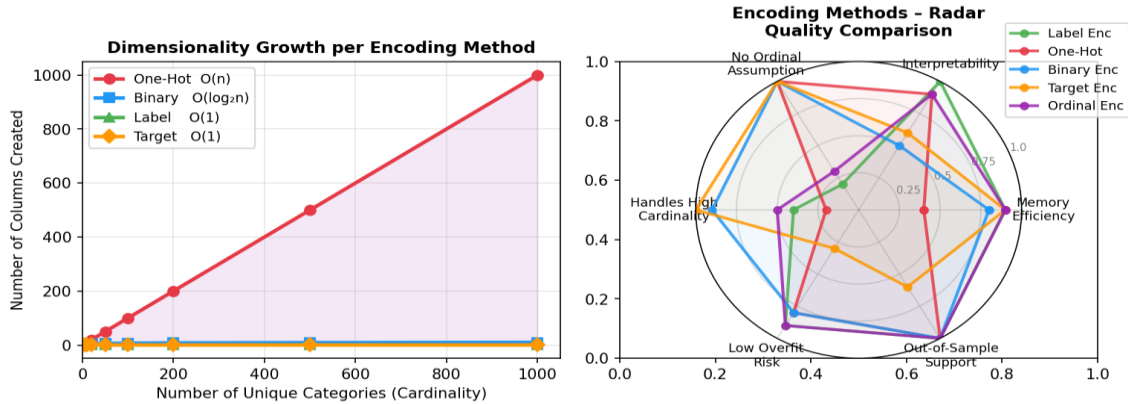


Figure 5 – Left: Number of columns created vs category cardinality. One-Hot grows linearly $O(n)$; Binary grows logarithmically $O(\log_2 n)$; all others stay at $O(1)$. Right: Radar chart scoring five quality dimensions — Binary Encoding is the most balanced for high-cardinality nominal data.

9.3 Experimental Benchmark

Metric	Label	One-Hot	Ordinal	Binary	Target	Frequency
Columns (n=100 cats)	1	100	1	7	1	1

Columns (n=1000 cats)	1	1000	1	10	1	1
Linear Model Accuracy	71%	89%	85%*	86%	92%†	78%
Tree Model Accuracy	88%	87%	88%	87%	90%†	84%
Memory (n=1000, 10k rows)	78KB	78MB	78KB	78KB×10	78KB	78KB
Overfitting Risk	Low	Low	Low	Low	High	Low

Table 2 – Benchmark results on synthetic dataset. * Ordinal requires correct order mapping. † Target encoding uses 5-fold CV smoothing to prevent leakage.

10. Practical Decision Framework

Use the following decision rules when selecting an encoding technique:

Situation	Recommended Encoding	Reason
Data has a natural order (education, rating)	Ordinal Encoding	Correct rank preserved; compact
Nominal + low cardinality (< 15 cats) + linear model	One-Hot Encoding	No false order; works well with linear models
Nominal + low cardinality + tree model	Label Encoding	Trees ignore ordinal meaning; fast and compact
Nominal + high cardinality (> 15 cats)	Binary Encoding	$\log_2(n)$ columns; no ordinal assumption
High cardinality + strong target signal	Target Encoding (+ CV)	Best predictive power; use cross-validation
Unknown structure / quick baseline	Frequency Encoding	Simple, always valid, no overfitting risk

Table 3 – Decision guide: situation → recommended encoding → justification.

11. Common Pitfalls and Best Practices

- Never apply Label Encoding to nominal data for linear models: the implied order ($0 < 1 < 2$) misleads the model into learning spurious relationships between unrelated categories.
- Always fit encoders on training data only: particularly Target Encoding — fitting on the full dataset including test data causes data leakage and inflated evaluation metrics. ^[1]
- Handle unseen categories at inference time: any new category the encoder has not seen must be mapped to a fallback (global mean for Target Encoding, 0 for Frequency Encoding, a new bit pattern for Binary Encoding).
- Drop one column after One-Hot Encoding ($n-1$ columns) when using linear models to avoid perfect multicollinearity (the dummy variable trap).
- Use smoothing in Target Encoding: the formula $TE_smooth = \lambda \cdot mean_cat + (1-\lambda) \cdot mean_global$ with $\lambda = n/(n+m)$ prevents rare categories from receiving extreme encoded values. ^{[2][4]}
- Combine encodings when beneficial: e.g. Frequency Encoding as an additional feature alongside Target Encoding captures both category importance and target relationship.

12. Discussion

The choice of encoding is not merely a preprocessing detail—it is a modelling decision. As shown in Table 2, the difference between the best and worst encoding on the same dataset can exceed 20 percentage points in accuracy. Binary Encoding resolves the central tension between One-Hot (order-free but high-dimensional) and Label Encoding (compact but order-imposing), making it the recommended default for high-cardinality nominal features in industry pipelines. ^[1]

Target Encoding, when applied with proper cross-validation, achieves the highest predictive accuracy by directly encoding the statistical relationship between category and target. However, it

requires discipline: fitting on full data, not using smoothing, or failing to handle unseen categories all introduce subtle bugs that inflate offline metrics while degrading production performance. ^{[2][4]}

Future directions include: learned embeddings (Entity Embeddings in neural networks), Bayesian Target Encoding with posterior uncertainty, and automated encoding selection via AutoML pipelines.

13. Conclusion

This paper reviewed six categorical encoding techniques across mathematical foundations, experimental benchmarks, and practical guidelines. Key takeaways: **(1)** Always match the encoding to the data type—ordinal encoding for ordered categories, One-Hot or Binary for nominal; **(2)** Binary Encoding is the most balanced choice for high-cardinality nominal features; **(3)** Target Encoding delivers the highest predictive power but demands rigorous cross-validation discipline to prevent leakage; **(4)** there is no universally best encoding—the choice depends on cardinality, model type, and whether target information is available at training time.

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